

Mechanisms of Autoimmunity in Rheumatologic Disease

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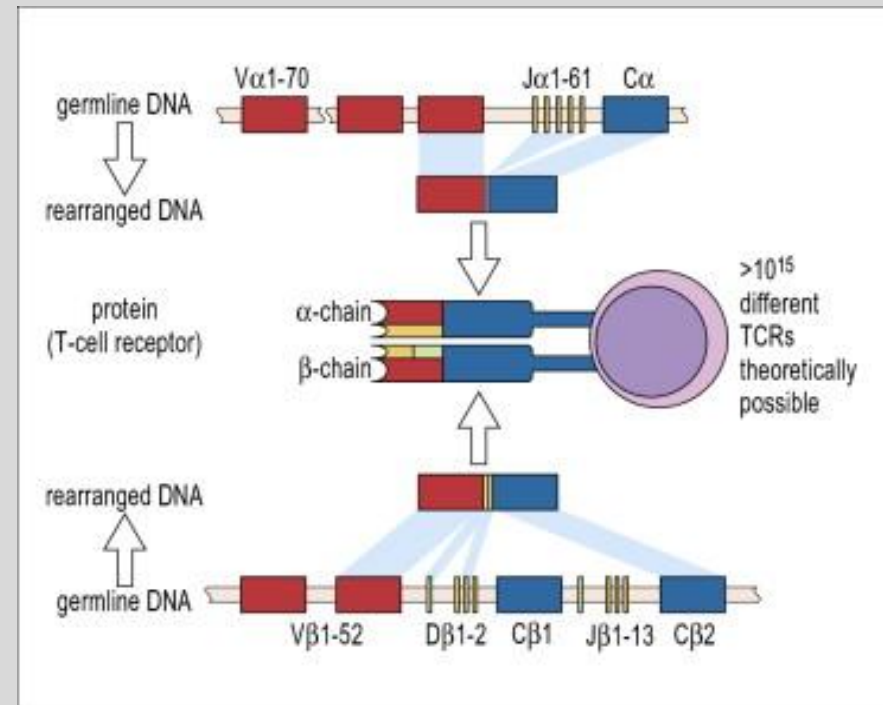
Session Objectives

1. Describe the phenomena of tolerance in immunity and the pathologic implications
2. Compare central to peripheral tolerance
3. Define autoimmunity
4. Describe the three stages of an autoimmune disease
5. Identify the mechanisms of pathogenesis of autoimmune diseases

- **Fulfill two contradictory requirements:**

1. Large variety of different receptors to avoid 'holes' that could be exploited by pathogens
2. the receptor repertoire must be shaped to prevent the immune system from attacking the host

Generation of T-cell receptor (TCR)



Source: David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013.

Immunology 8th edition. Elsevier publisher. ISBN: 978-0-323-08058-3

figure 19.1.

Origins of Autoimmunity

- Paul Ehrlich coined the term *horror autotoxicus* for the necessity to avoid immunological reactions against self-antigens

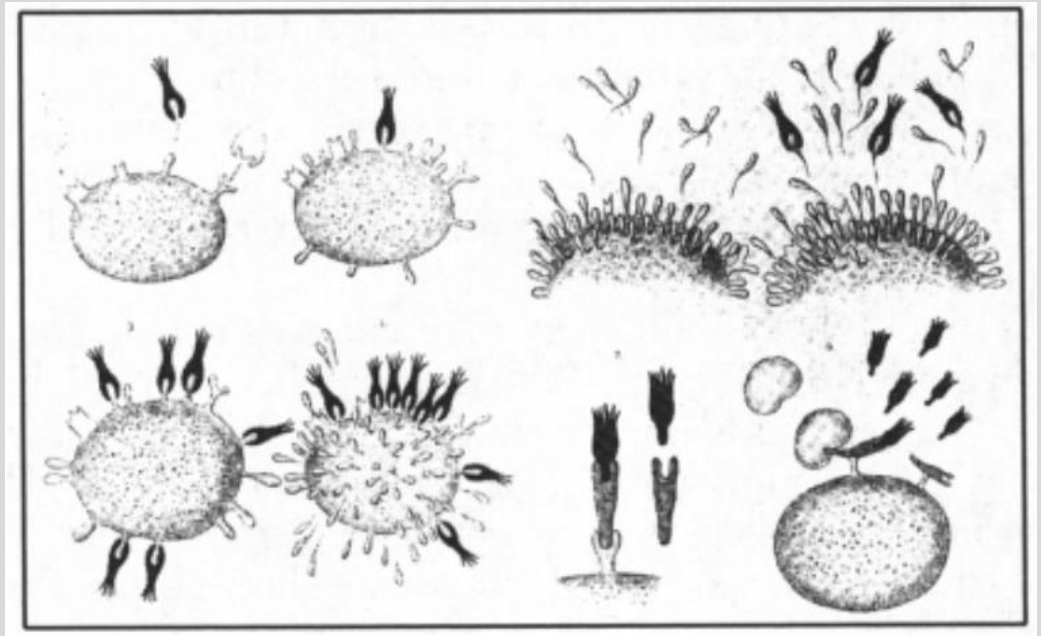


Illustration of Ehrlich's Side Chain Theory

Source: Travis AS. *History and Philosophy of the Life Sciences*
Vol. 30, No. 1 (2008), pp. 79-97

- **Antigen-specific unresponsiveness**
- Result of a programmed developmental process
- Adaptive IR learns to tolerate:
 1. self-antigens (autoantigens)
 2. exogenous antigens (dietary proteins)
- Autoimmunity requires the presence of self-reactive CD4⁺ T-cells (*exception*)

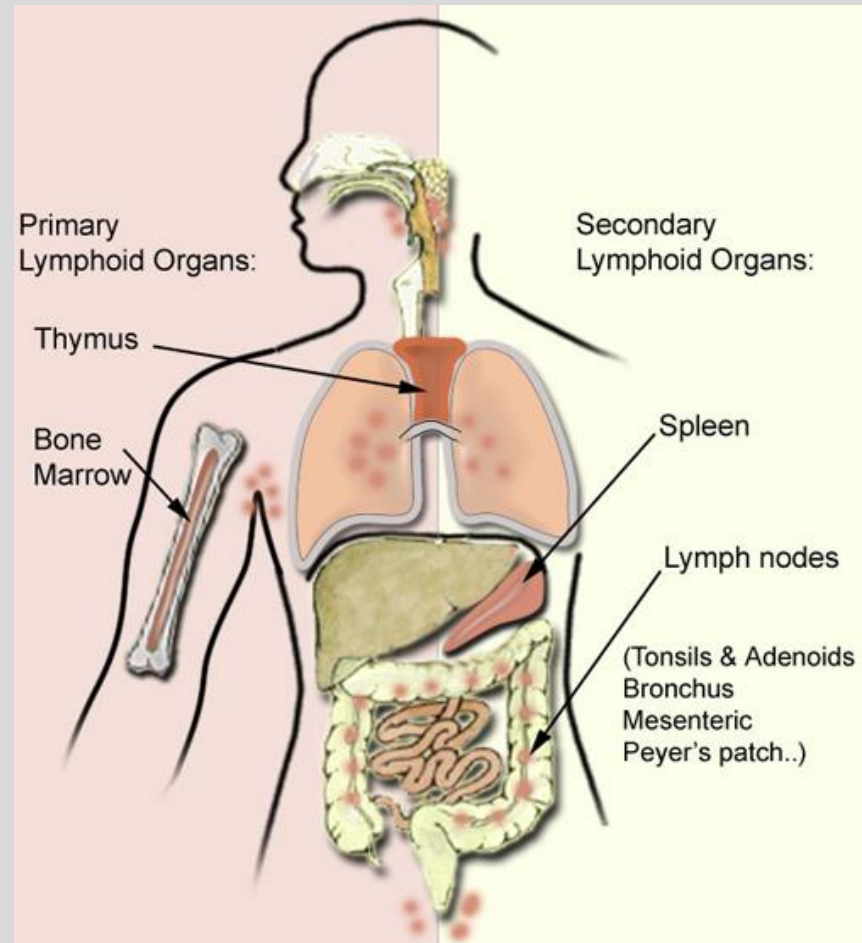
Types of Tolerance

Central Tolerance

- Primary lymphoid organs
 - Bone marrow → B Cells
 - Thymus → T Cells
- Important role but only part of the story

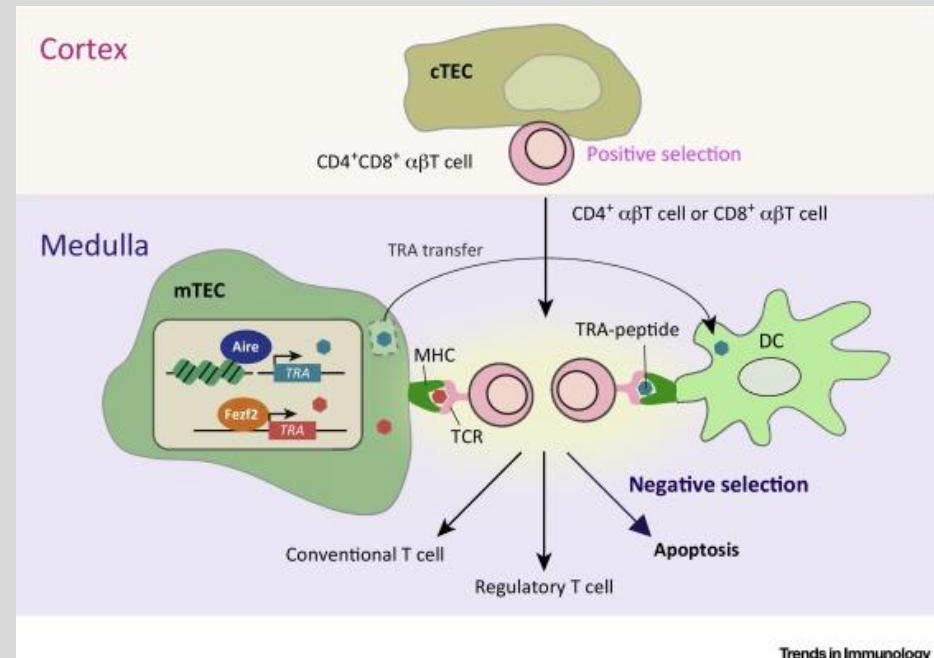
Peripheral Tolerance

- Secondary lymphoid organs



Central Tolerance | Thymus | Site of T-cell “Education”

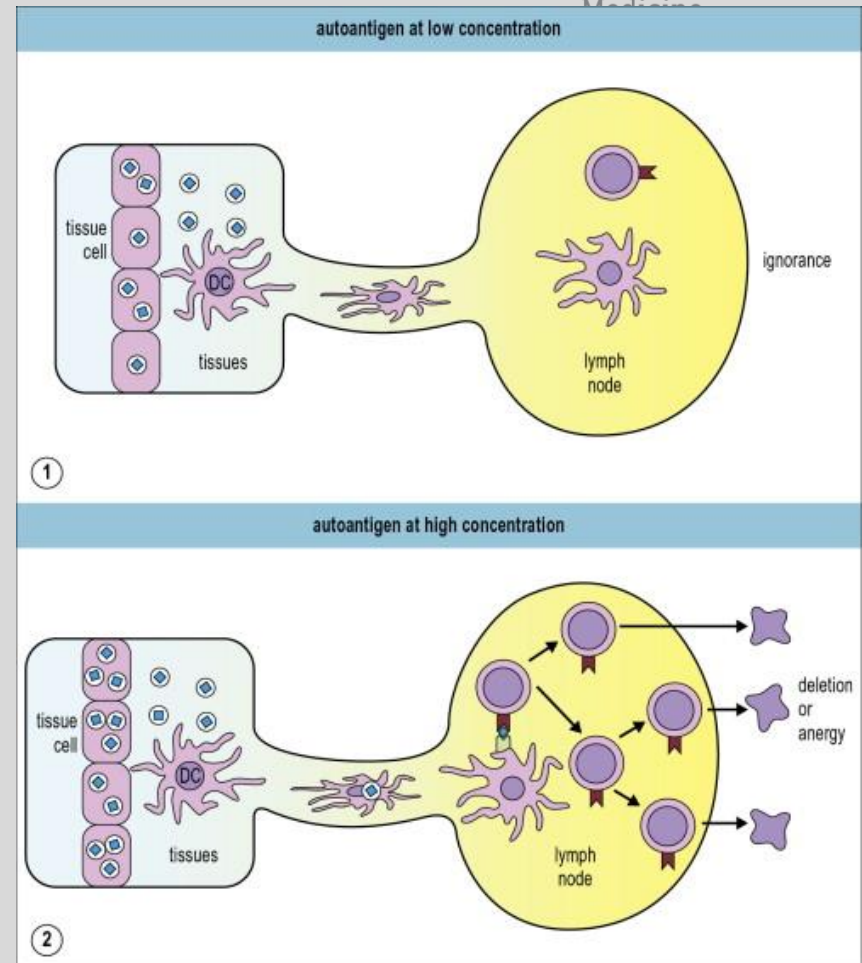
- Thymocytes: T-Cell precursors
- **Positive Selection**
 - Process of selecting thymocytes with receptors that can both recognize & bind MHC I receptors
- **Negative Selection**
 - Deletion of T-cells that bind too avidly to MHC receptors or self-peptides presented on MHC receptors
- < 5% of total T-cells survive → exit as naïve mature T cells



Source: Takaba H. The Mechanisms of T Cell Selection in the Thymus.
[https://www.cell.com/trends/immunology/fulltext/S1471-4906\(17\)30148-5](https://www.cell.com/trends/immunology/fulltext/S1471-4906(17)30148-5)

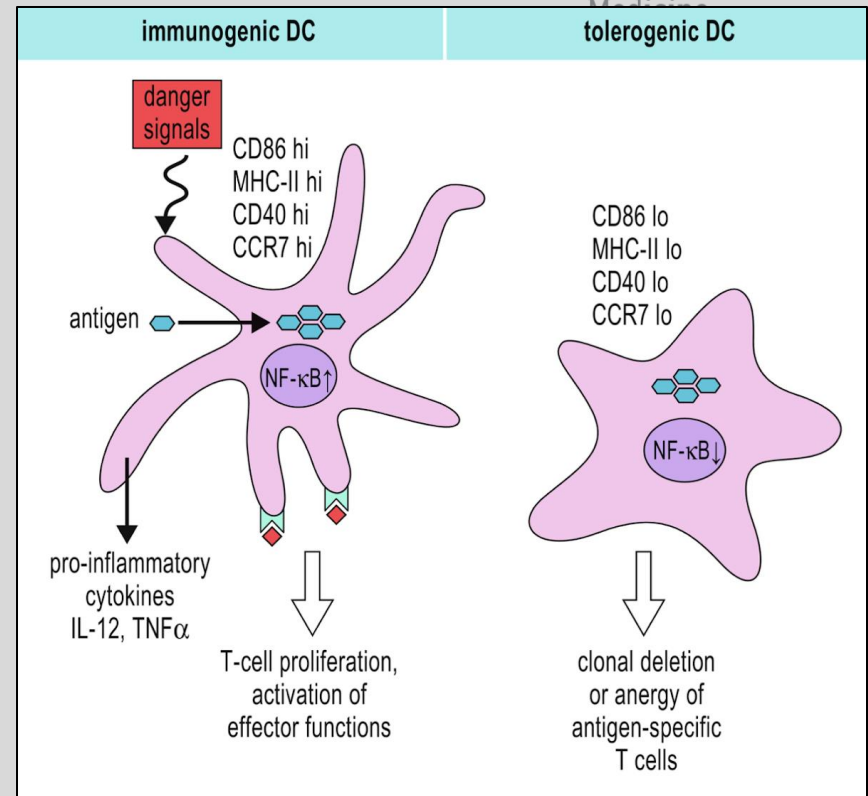
Central Tolerance | T-cells

1. 30% of low-avidity self-reactive T-cells clones → periphery
2. Thymic deletion of autoreactive T-cells is not the only mechanism responsible for self-tolerance → peripheral tolerance
3. Antigen dose and TCR avidity
4. Medium to high affinity self reactive T-cells become T_{reg}



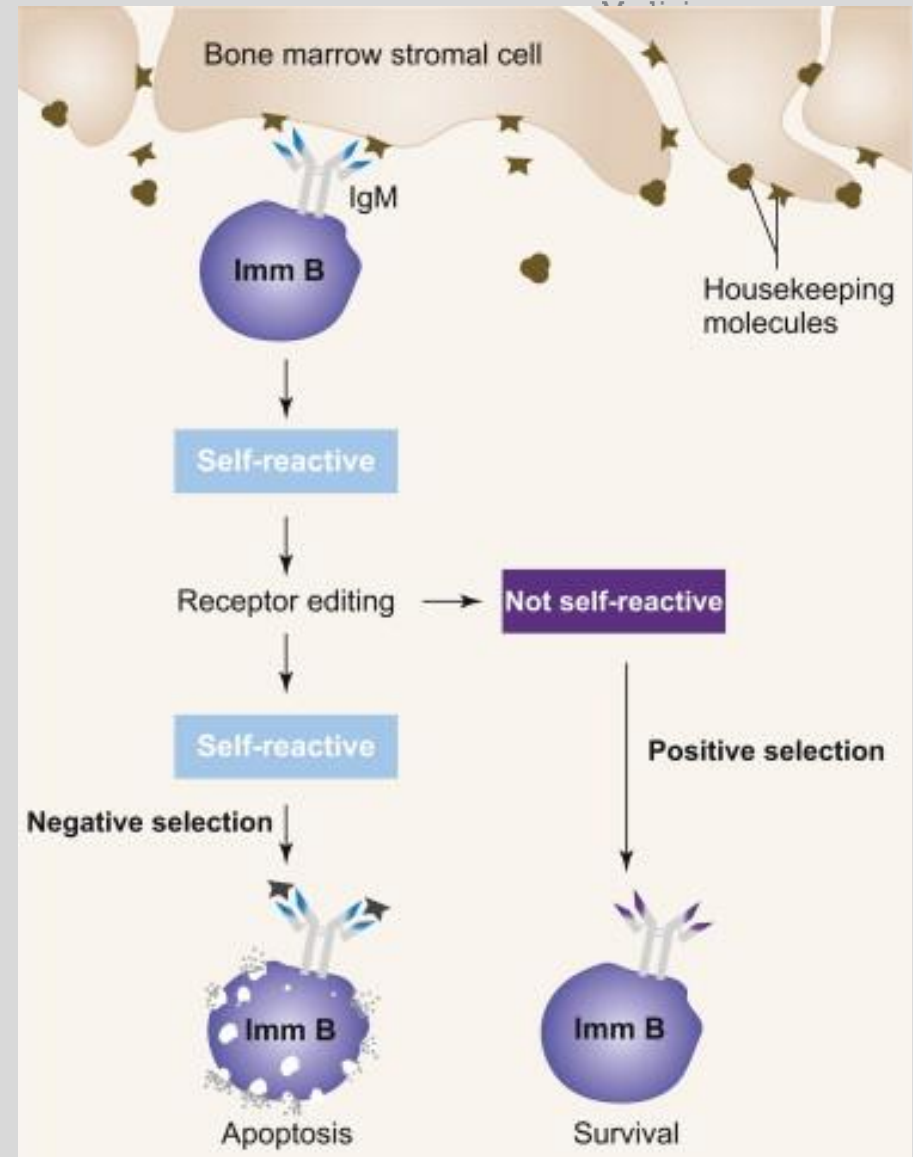
Source: David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. Immunology 8th edition. Elsevier publisher. ISBN: 978-0-323-08058-3 Fig 19.w1

1. Autoreactive T-cells are part of the **normal repertoire** (*e.g., T-cells that recognize insulin or myelin can be isolated from people without diabetes or MS*)
2. **Tolerogenic Dendritic cells** (*not immunogenic*)

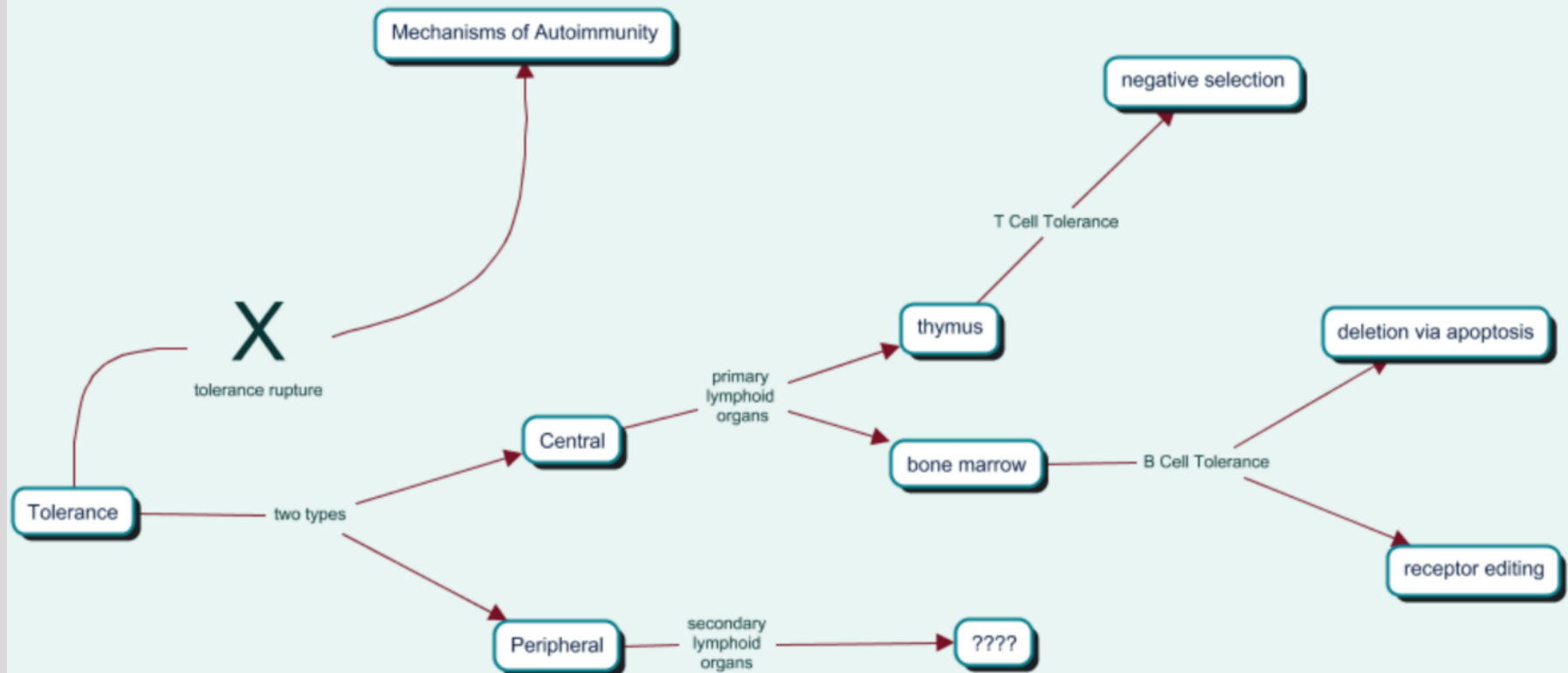


Source: Figure 11.7 from Chapter 11 in D. Male, R. Stokes Peebles, and V. Male. 2021. *Immunology* 9th edition. Elsevier publisher. ISBN: 978-0-7020-7844-6

- **Three major differences from T-cell tolerance:**
 1. **Occurs in the bone marrow:** Negative selection of B-cells strongly reactive with self-antigens → apoptosis
 2. **BCR Editing:** modification of light chain sequence (20 to 50%)
 3. Follow guidance from T-cells



Central Tolerance



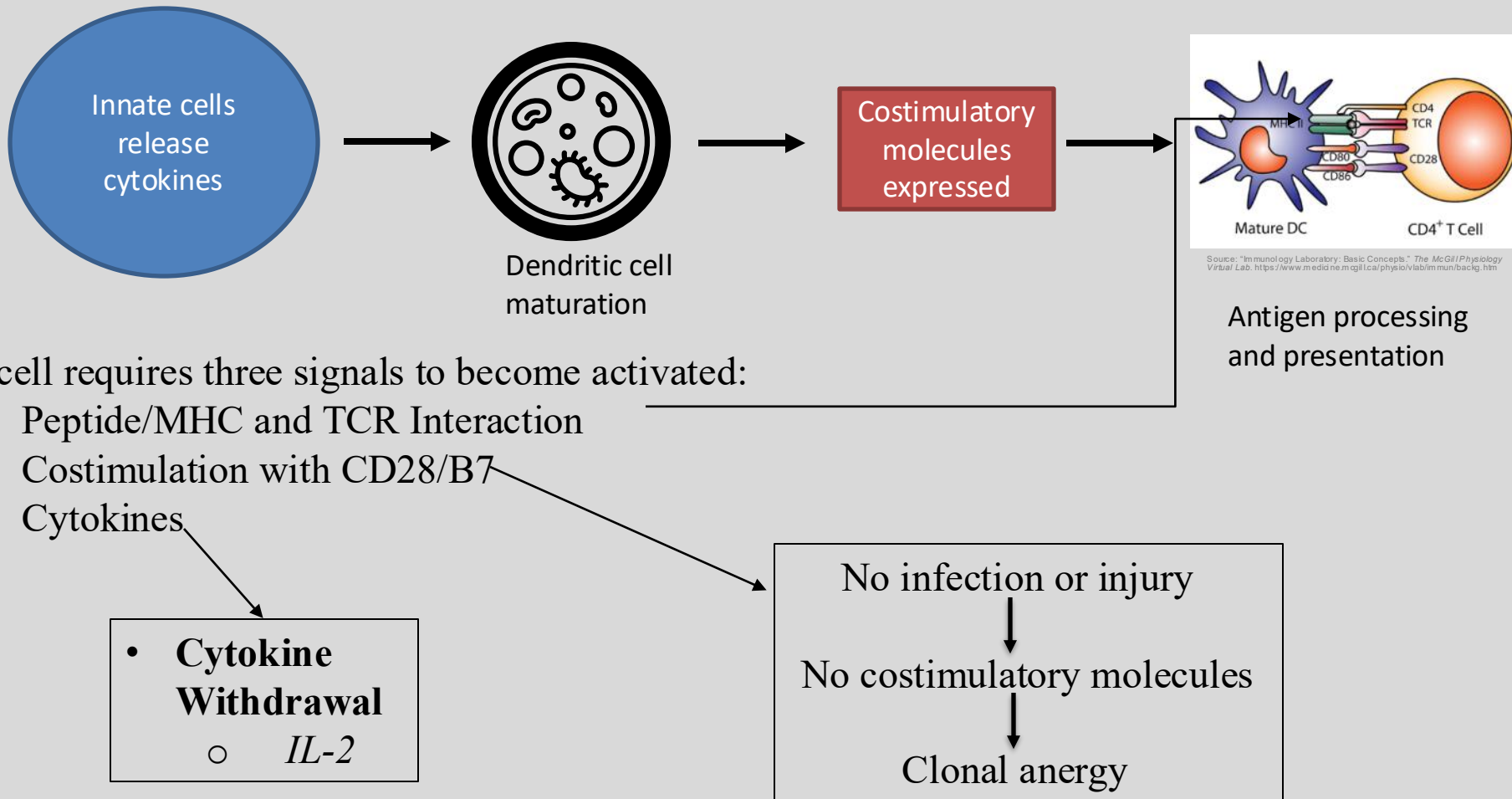
Mechanism of Peripheral Tolerance

1. Clonal Anergy
2. Immunologic Ignorance
3. Cytokine Withdrawal
4. Sequestration
5. Idiotypic Network Theory
6. T_{reg} cells

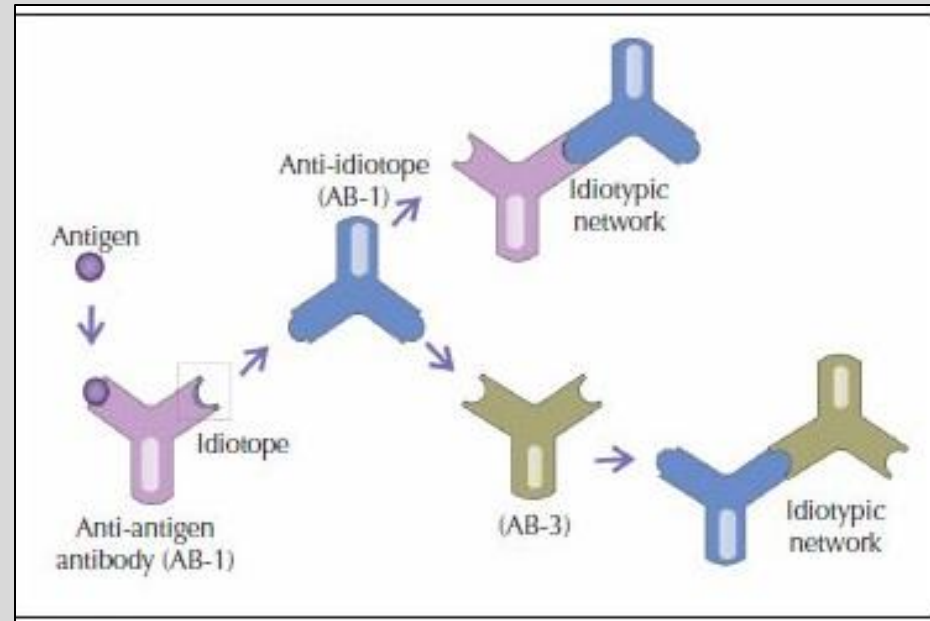
Peripheral Tolerance

- **Clonal anergy:** self-reactive T-cells may not receive the necessary costimulatory signals (CD28/B7)

- **Immunologic Ignorance**



- **Sequestration** of potentially harmful T-cells
 - Physical: BBB
 - Chemical: migration controlled by chemokines, selectins, & their receptors
- **Idiotype Network Theory:** antibodies directed against the self-reactive antibody



Souza, Alexandre Wagner Silva de, Mesquita Júnior, Danilo, Araújo, Júlio Antônio Pereira, Catelan, Tânia Tieko Takao, Cruvinel, Wilson de Melo, Andrade, Luís Eduardo Coelho, & Silva, Neusa Pereira da. (2010). Immune system: part III. The delicate balance of the immune system between tolerance and autoimmunity. *Revista Brasileira de Reumatologia*, 50(6), 665-679. <https://dx.doi.org/10.1590/S0482-50042010000600007> Fig 6

Peripheral Tolerance | T_{reg} cells

Medium → high affinity
self reactive T-cells

FOXP3: regulatory gene for
development and function

T_{reg} cell

identity

10% of all CD4+ T-cells

Selectively express CD3 receptor

Prevent and
suppress immune
responses

Cytotoxic to:

- CD8+ T cells
- NK cells
- B-cells
- DCs

Depletion
of IL-2

Secretion of

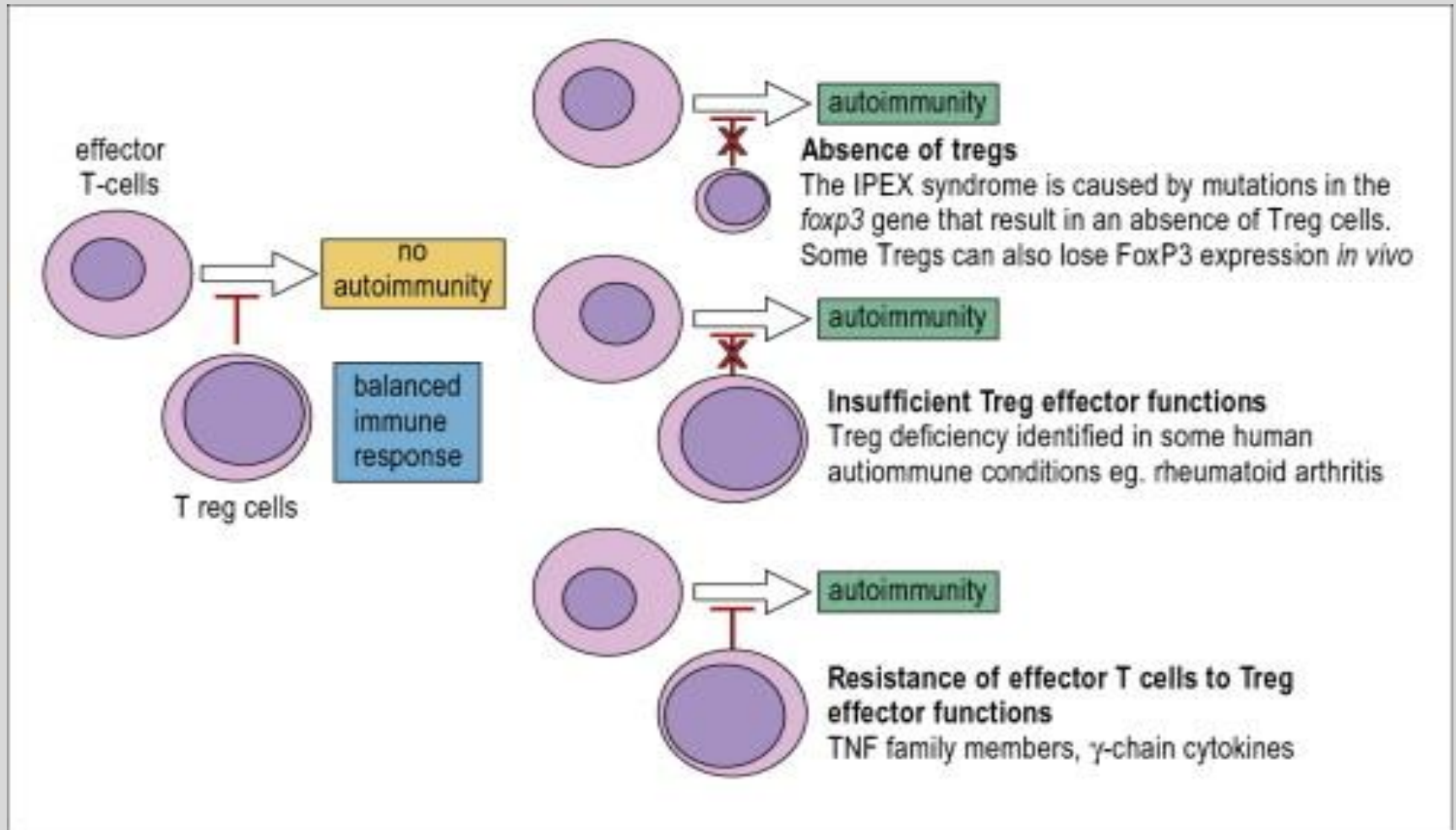
IL-10

TGF- β

IL-35

Importance:

- Dampen responses against microbial antigens, allergens, tumors, & allografts, and protect fetuses during pregnancy
- IPEX (immunodysregulation, polyendocrinopathy, enteropathy, X-linked) syndrome
- Autoimmunity due to loss of function or effectiveness (effector T cells become resistant to their action)

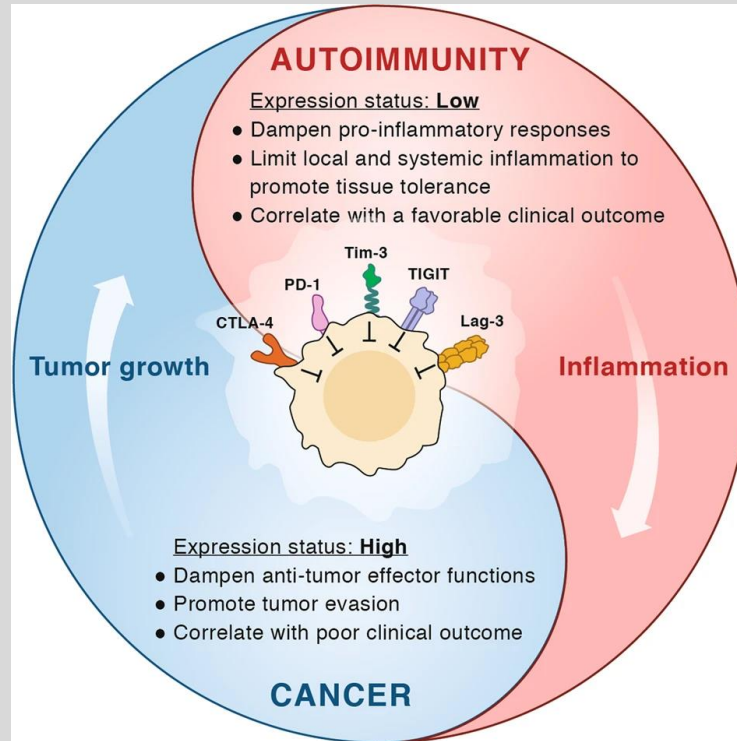


Source: David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013.

Immunology 8th edition. Elsevier publisher. ISBN: 978-0-323-08058-3 Fig

Establishing Peripheral Tolerance

- Major goal for innovative treatments of:
 - Autoimmunity
 - Allergy
 - Transplantation



Source: Schnell A, Bod L, Madi A, and Kuchroo VK. "The yin and yang of co-inhibitory receptors: toward anti-tumor immunity without autoimmunity." *Nature*. <https://www.nature.com/articles/s41422-020-0277-x/figures/1>

Overcoming Peripheral Tolerance

- Major goal for innovative treatment of:
 - Neoplasms

Review Questions

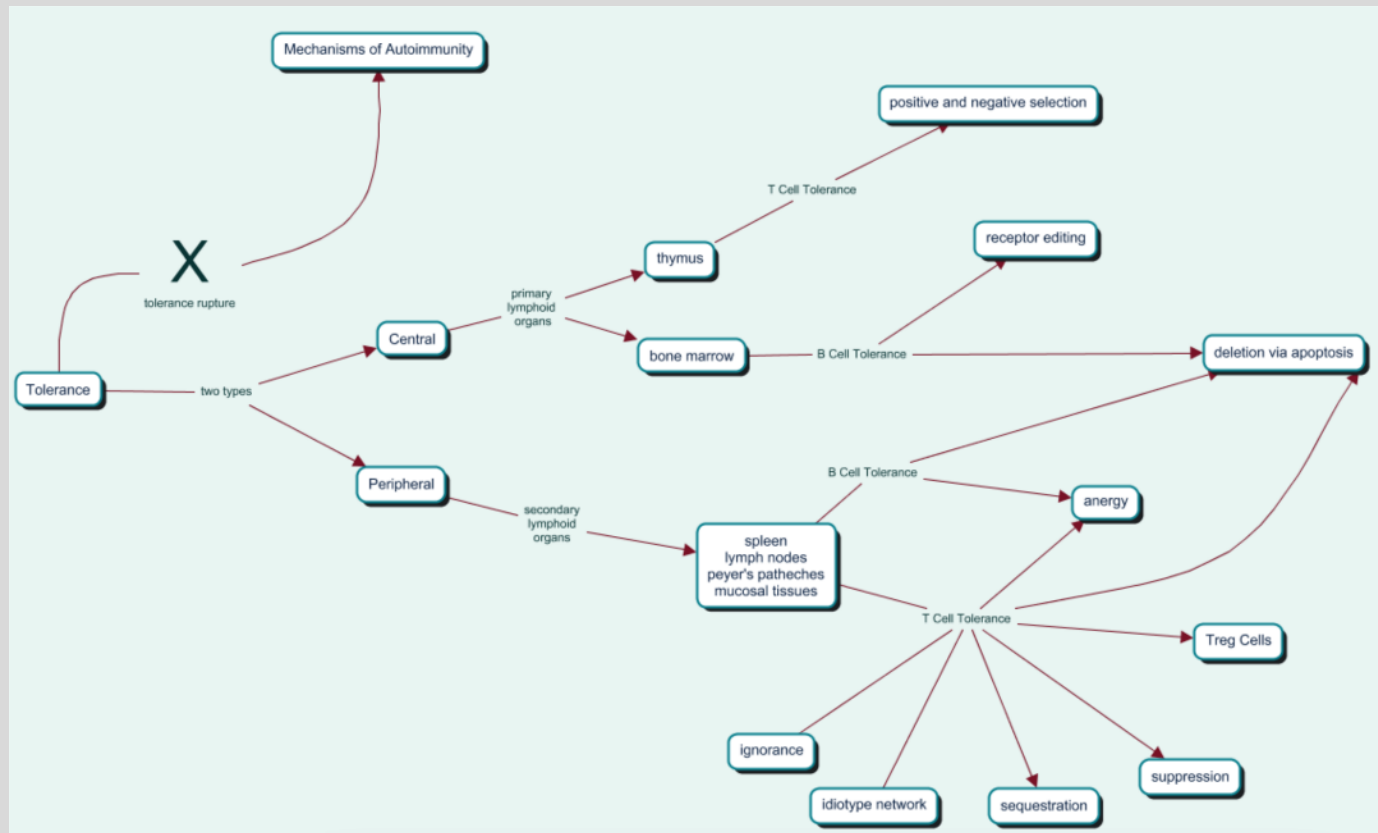
1. Which of the following is unique to central B-cell tolerance?
 - A. Immunologic Ignorance
 - B. Receptor Editing
 - C. Anergy
 - D. Apoptosis

2. Which of the following would not be used as a marker for T_{reg} cells?
 - A. CD25+
 - B. FOXP3
 - C. CD3+
 - D. CD8+

3. A mutation in the FOXP3 gene would directly affect which of the following cell types?
 - A. T-helper cells
 - B. NK cells
 - C. T-regulatory cells
 - D. Dendritic cells

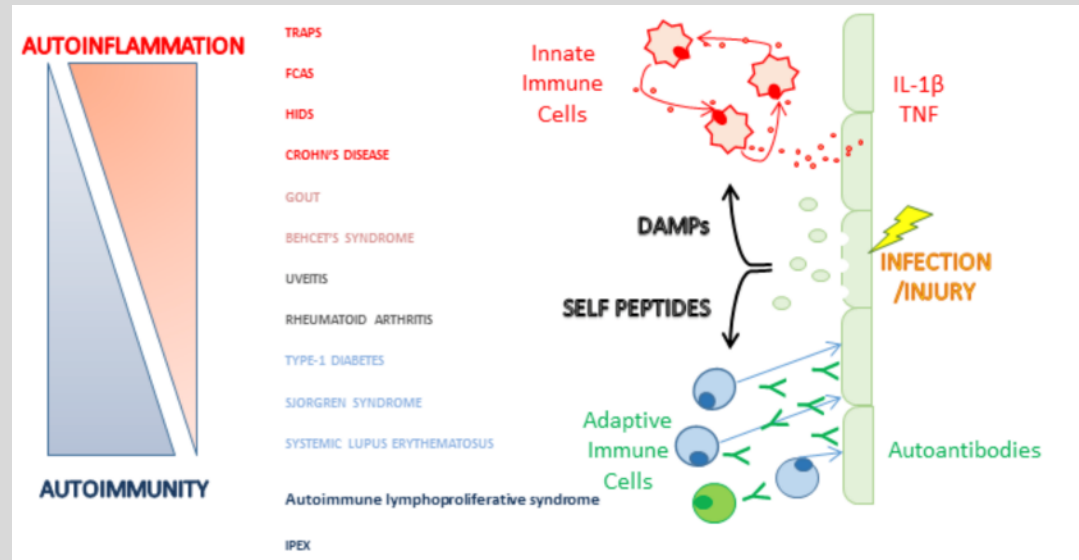
Definition of Autoimmunity

- Autoimmunity occurs when:
 - Self-tolerance broken (*tolerance rupture*) and an individual becomes victim of their own immune response
 - Loss of previously established tolerance



Autoimmunity $\neq$ Autoinflammation

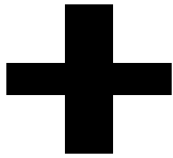
- ***Not*** a collateral damage from an inflammatory process or an autoinflammatory disease [e.g., TNF receptor-associated periodic syndrome (TRAPS)]



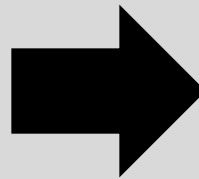
Source: Scrambler T. "Autoinflammation." *British Society for Immunology*.
<https://www.immunology.org/public-information/bitesized-immunology/immune-dysfunction/autoinflammation>

**Antigen driven process
characterized by:**

1. Specificity
2. High affinity
3. Memory



**Autoreactive
lymphocyte or
autoantibody**

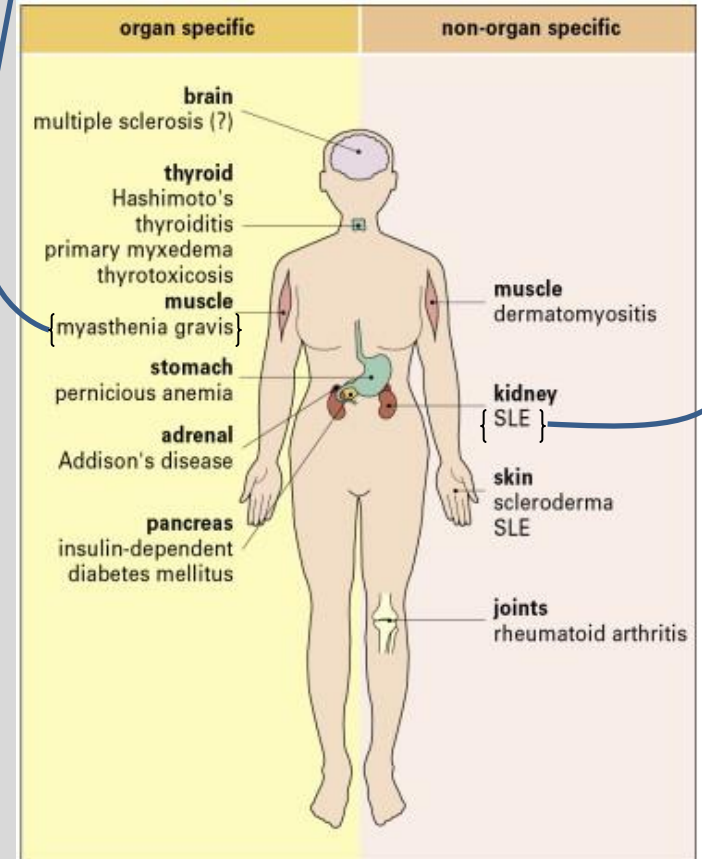
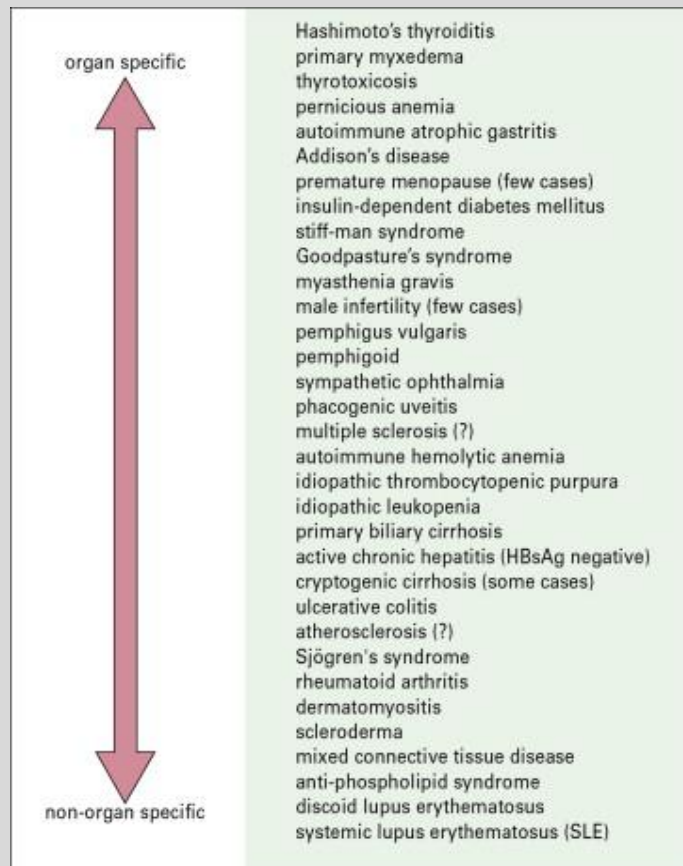


pathogenesis

recognition of an antigen by the
IS (either foreign or self) → assault
on its own self-antigens

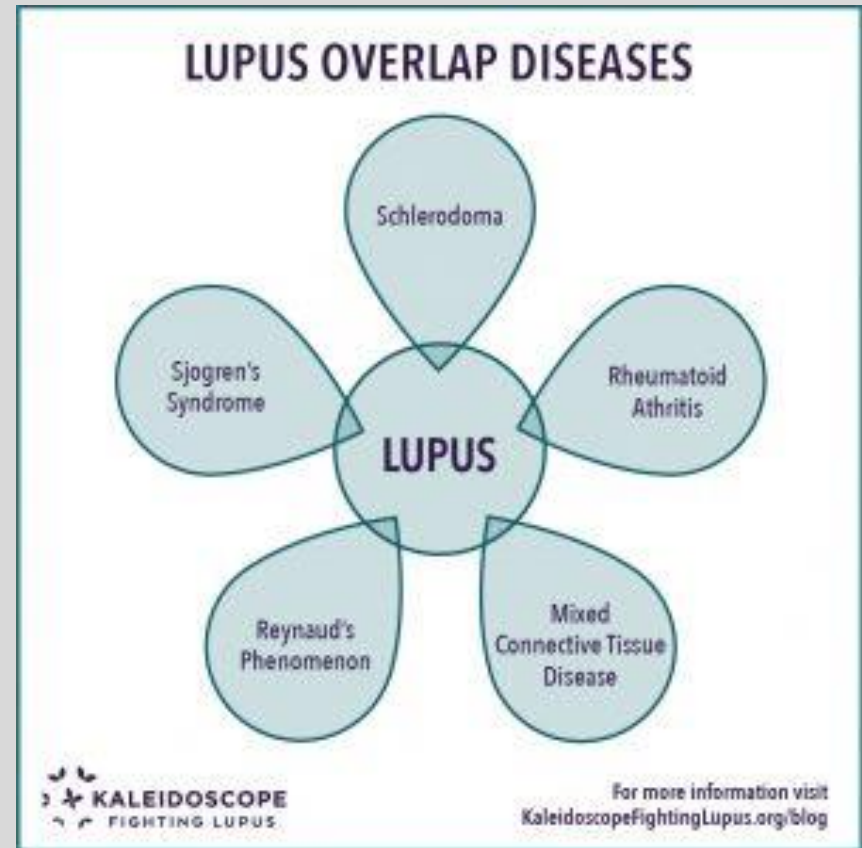
Autoimmune Diseases

Clinical Categories	Organ Specific	Systemic
Mechanism	Serum reacts to single autoantigen/restricted group of autoantigens in one organ	Serum reacts to multiple autoantigens



Autoimmune Diseases

- Autoimmune diseases
‘hunt in packs’
 - 30% of patients with SLE & Sjögren's syndrome have a second, third, or even fourth autoimmune disease
- Systemic autoimmune rheumatic diseases show considerable overlap



Source: Heintz L. Lupus Overlap Diseases and Overlap Syndrome. <https://kaleidoscopefightinglupus.org/lupus-overlap-diseases-and-overlap-syndrome/>

Stages of Autoimmune Diseases

Three Stages



Priming



Subclinical



Clinical

Stages of Autoimmune Diseases

Stage One: Priming

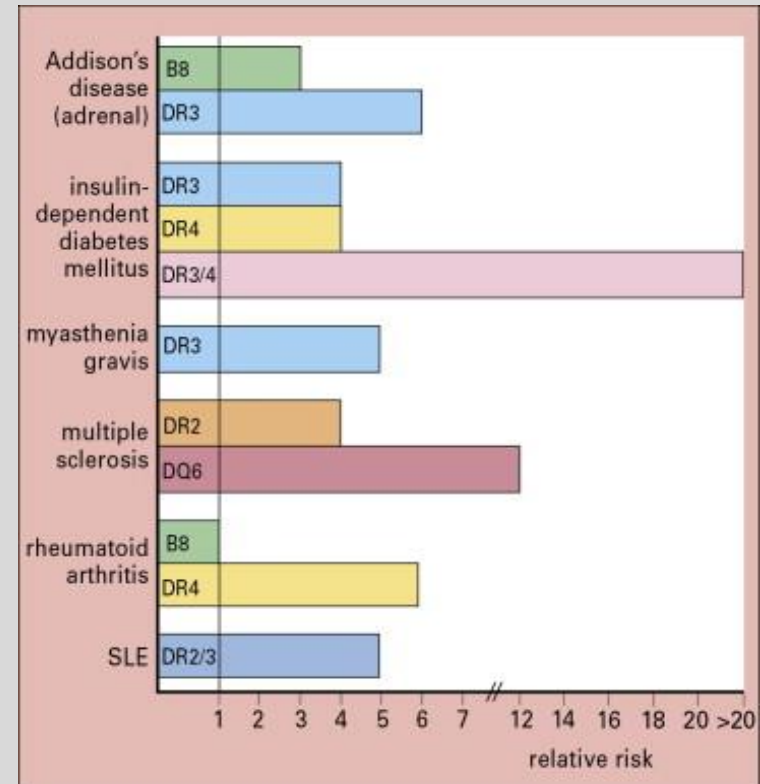
Genetic

genes that confer a
genetic risk

polygenic; HLA-B27

Environmental Exposure

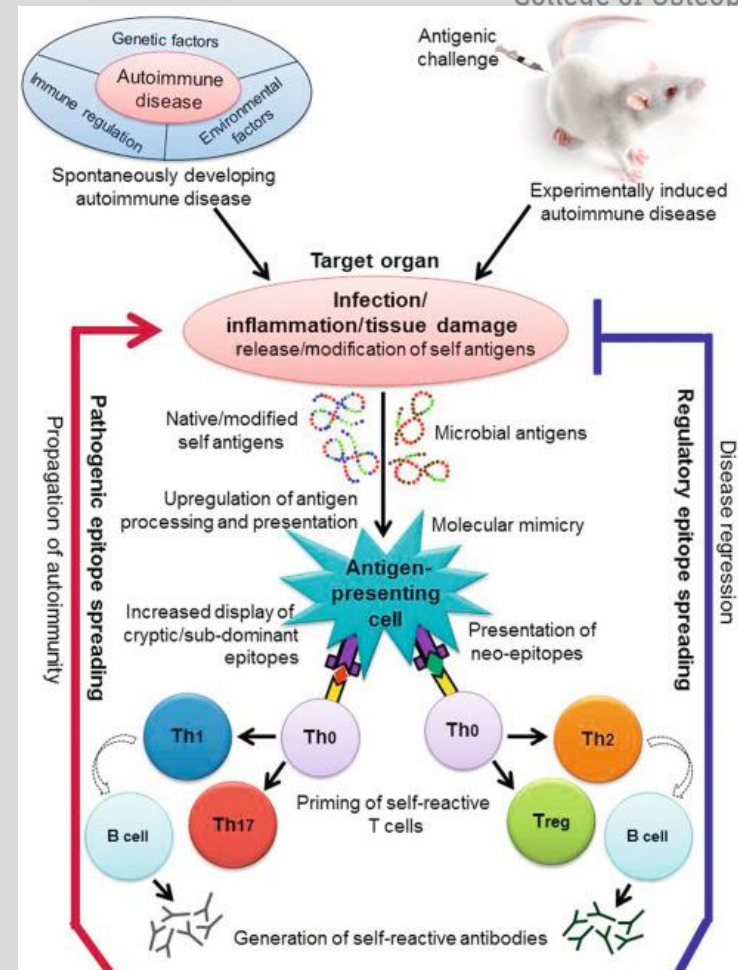
Infection or UV radiation



Source: Male D, Brostoff J, Roth D, and Roitt I (2013). Immunology 8th ed. Elsevier publisher. ISBN: 978-0-323-08058-3 Fig. 20.6

Stage Two: Subclinical

- Target organs have not become sufficiently damaged
- Leads to epitope spread



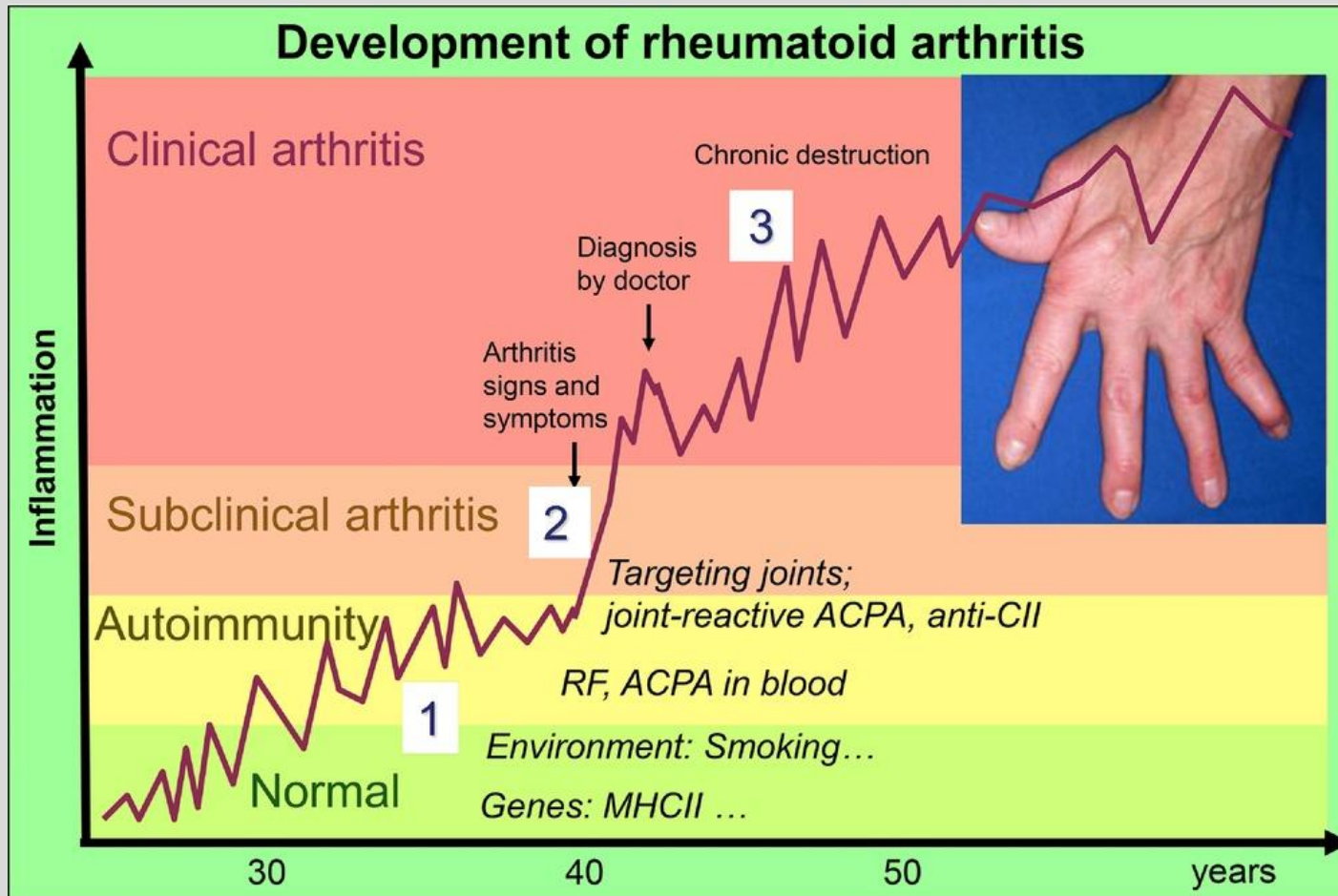
Source: Venkatesha SH, Durai M, and Moudgil KD (2015). Chapter 4: Epitope Spreading in Autoimmune Diseases. <https://www.sciencedirect.com/science/article/pii/B978044463269200039>

Stages of Autoimmune Diseases

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Stage Three: Clinical



Source: Holmdahl R, Malmstrom V, and Burkhardt H (2014). Autoimmune priming, tissue attack and chronic inflammation- The three stages of rheumatoid arthritis. <https://onlinelibrary.wiley.com/doi/full/10.1002/eji.201444486>

Review Questions

1. A certain autoimmune condition causes inflammation, vasculitis, and perifascicular atrophy. Is this an organ specific or systemic autoimmune disease? _____.

2. Which of the following is true of autoimmunity?
 - A. It involves the loss of tolerance
 - B. It occurs in any individual with autoreactive T-cells
 - C. An individual can only have one autoimmune disease
 - D. There are no known genetic risk factors

Mechanism of Pathogenesis of Autoimmune Diseases

Super Antigens

Defective Receptors

Molecular Mimicry

Defective Apoptosis

Epitope Spreading

Cryptic Epitope Exposure

Hapten Effect

Inappropriate MHC Expression and Upregulation

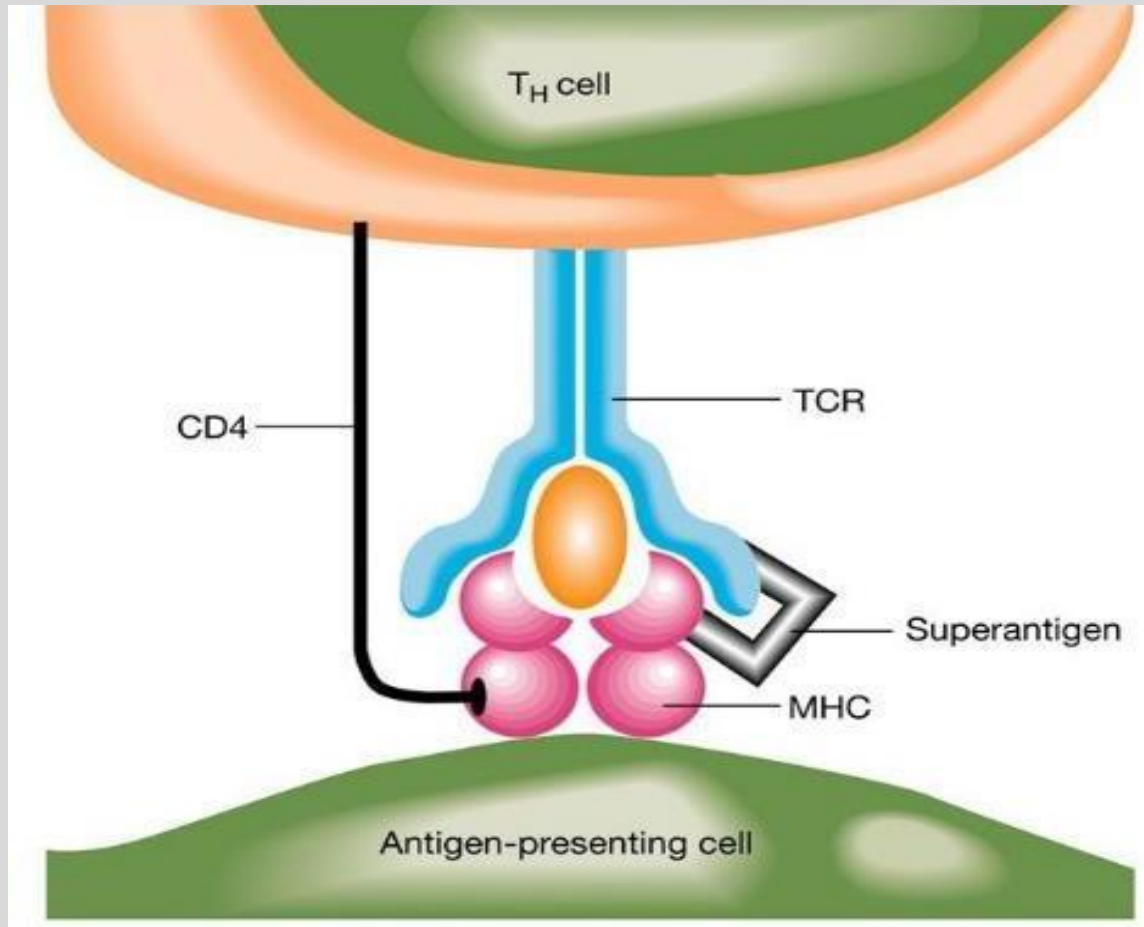
Defective T_{reg} Response

Hormonal Imbalances

Cytokine Dysregulation

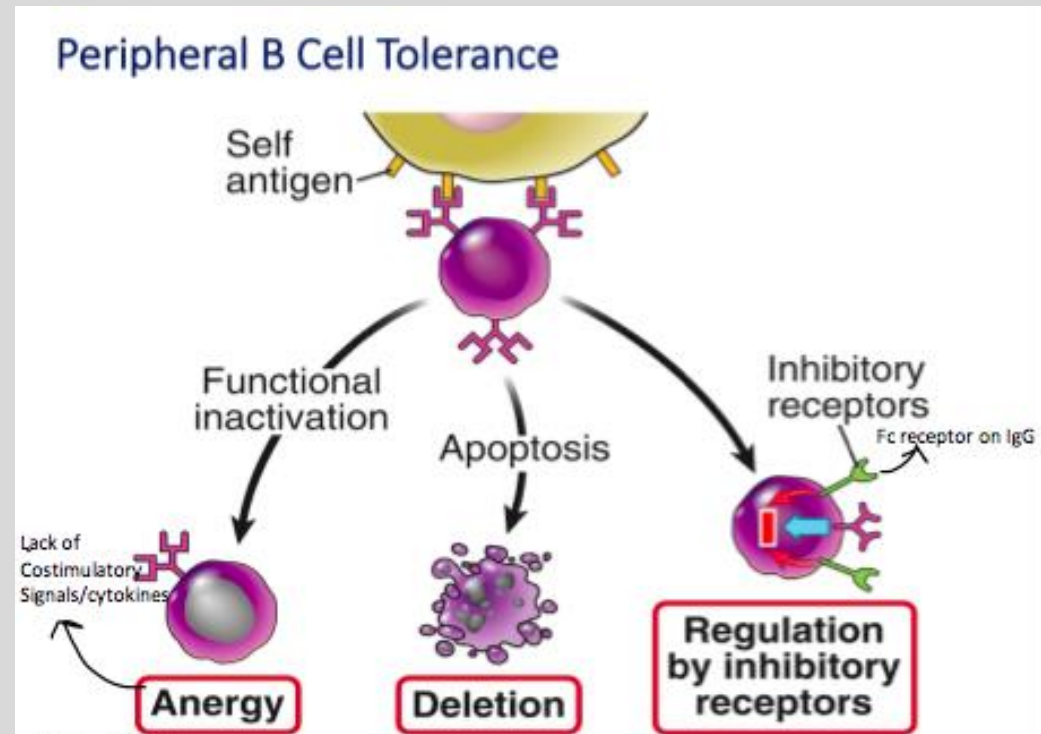
Pre-Existing Organ Defects

Superantigens



Source: http://images.slideplayer.com/18/5680536/slides/slide_42.jpg

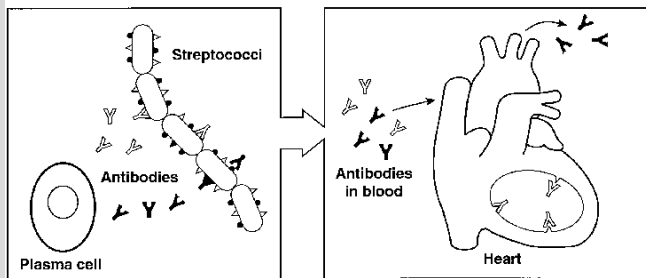
- **Autoreactive B-cells with defective receptors**
 - do not respond to inhibitory signals to maintain self-tolerance



Source: <https://memorang.com/flashcards/146143/Lecture+4+-+Autoimmunity>

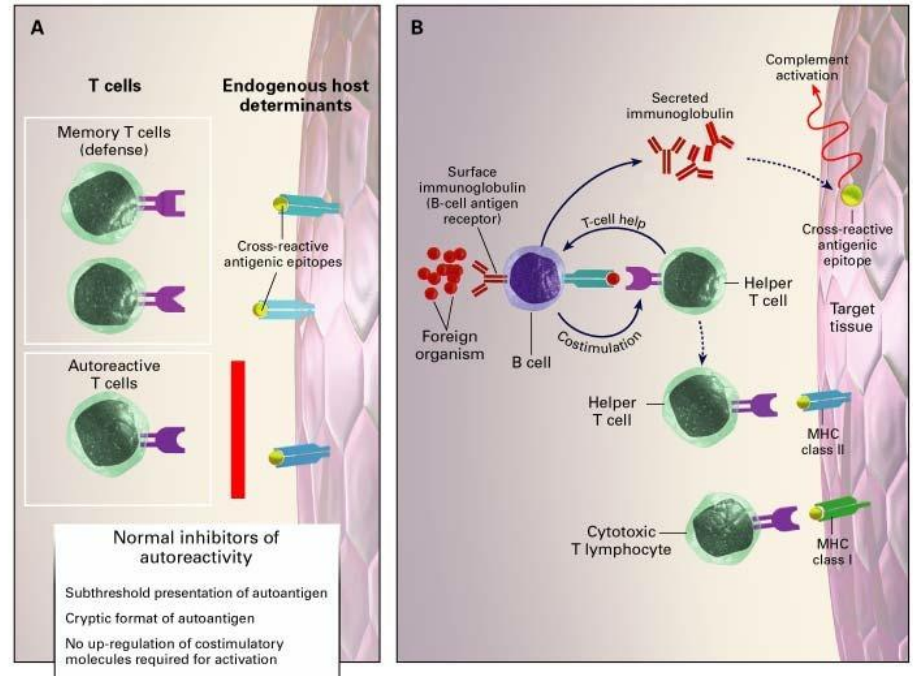
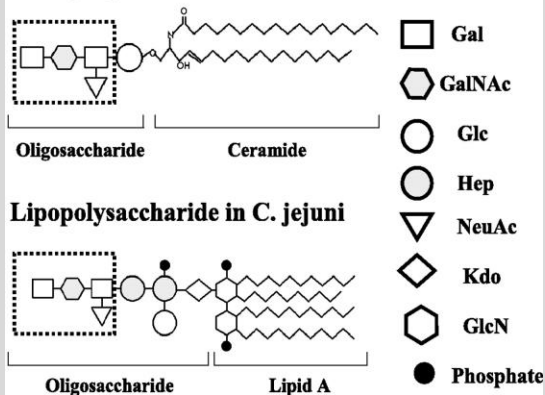
Molecular Mimicry

Rheumatic fever/*S. pyogenes*



Guillain–Barré Syndrome/*C. jejuni*

GM1 ganglioside in nerve cell membrane



TSH receptor/*Y. enterocolitica* homology

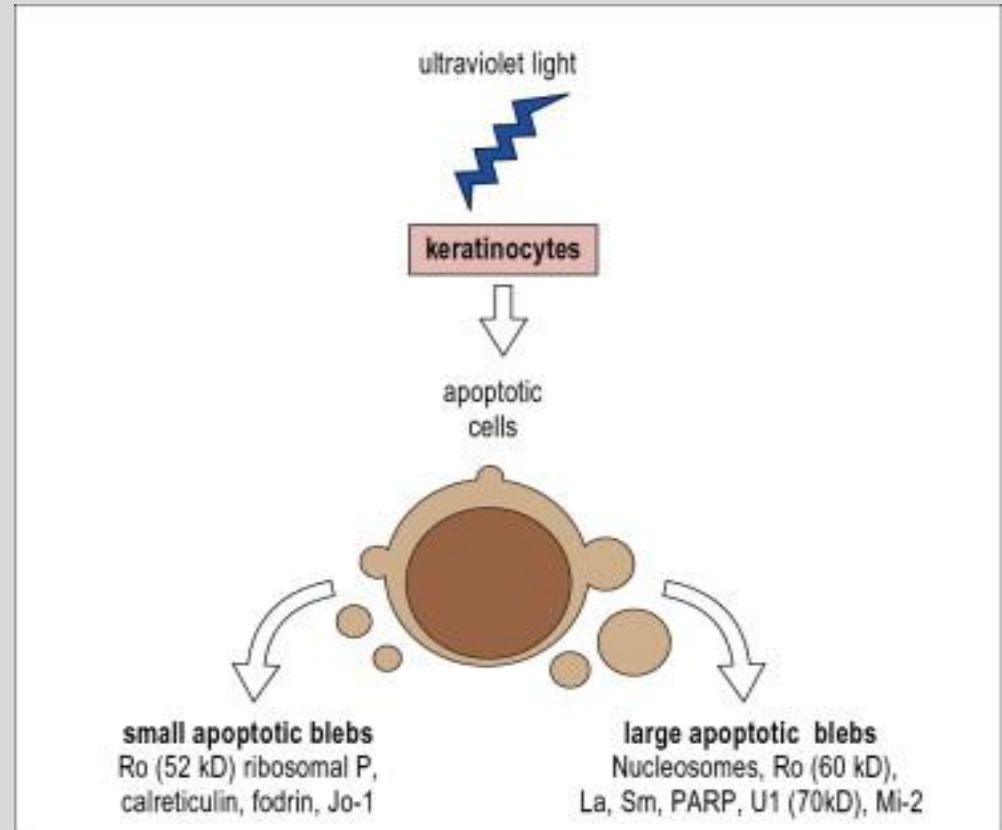
Source: Guillain-Barre Syndrome: <https://iaj.asm.org/content/74/12/6517>

Rheumatic Fever: https://www.researchgate.net/publication/11825642_Immunologic_tolerance/figures?lo=1

Molecular Mimicry: Source: <https://www.nejm.org/doi/full/10.1056/nejm199912303412707>

Defective Apoptosis:

- a) Accelerated apoptosis of cells with increased release of self-antigens
- b) Defective presentation of apoptotic cell antigens to lymphocytes
- c) Failure of clearance of apoptotic cells by $M\phi$ $\rightarrow$ prolonged exposure of internal autoAgs (SLE)

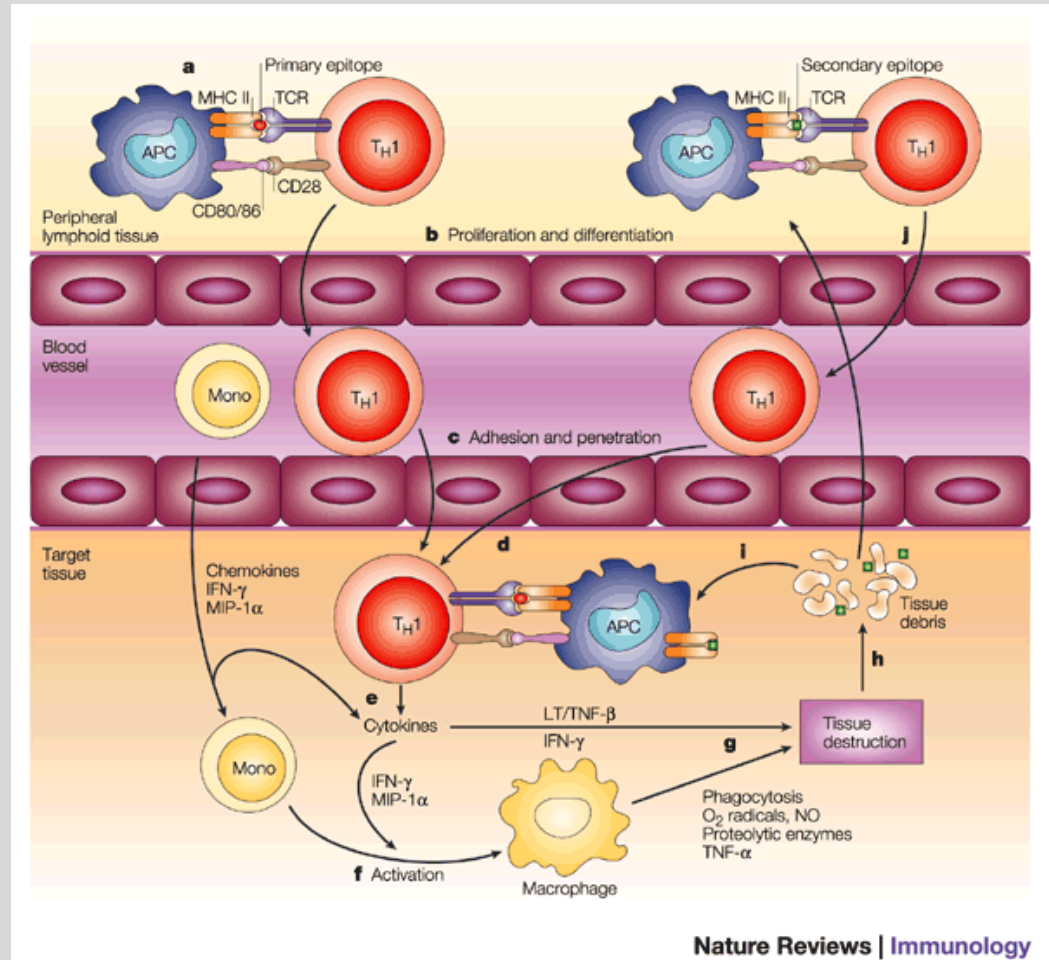


Source: David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. Immunology 8th edition. Elsevier publisher. ISBN: 978-0-323-08058-3 Fig 20.10

• Epitope Spreading

- switching from targeting the primary epitope to also targeting other epitopes

- *occurs after tissue/organ destruction, which exposes more epitopes*



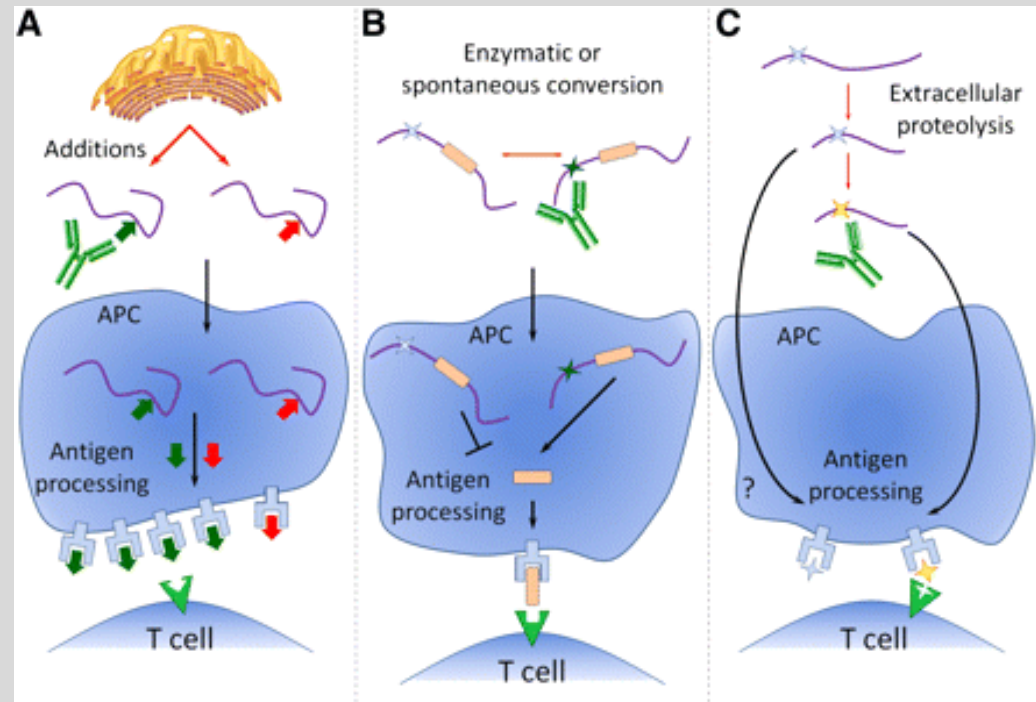
Source: Vanderlugt CL and Miller SD (2002). "Epitope spreading in immune-mediated diseases: implication for immunotherapy."
<https://www.nature.com/articles/nri724>

• Cryptic Epitope Exposure

○ exposure of a new self-antigen

a) type 1 diabetes mellitus

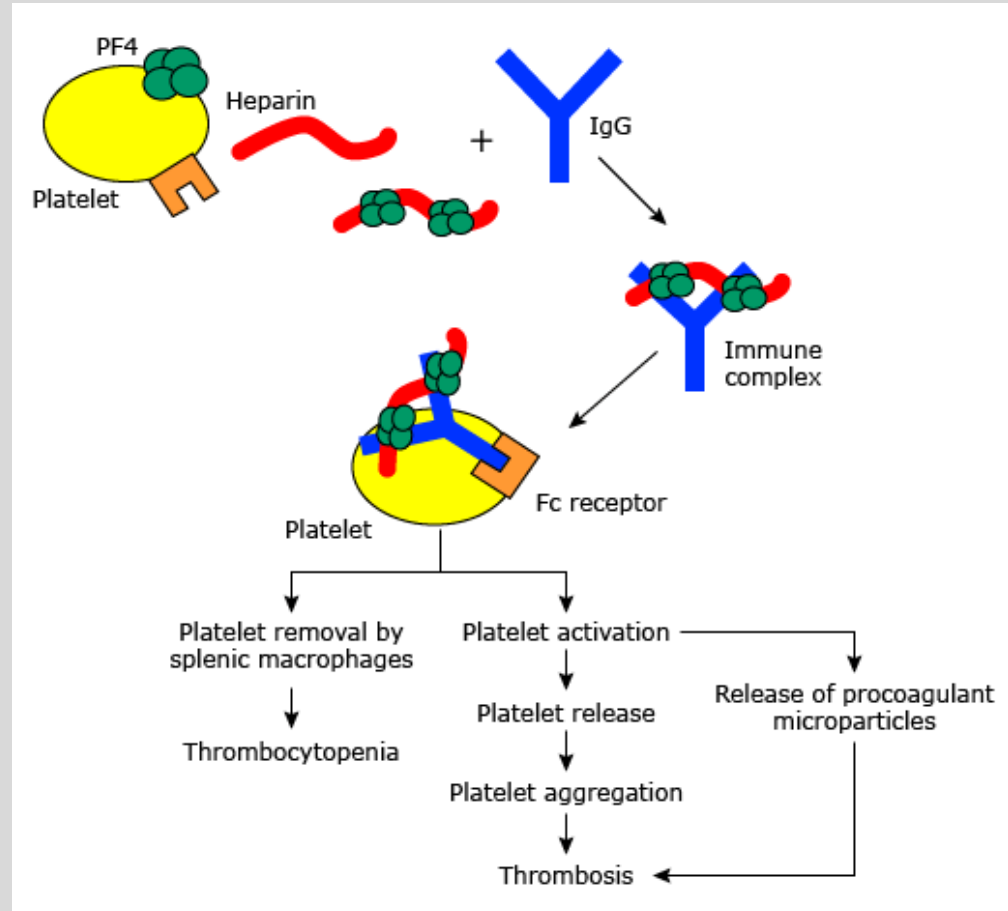
b) post-viral myocarditis:
Coxsackie virus must infect heart tissue



Source: Dunne JL, Overbergh L, Purcell AW, Mathieu C. Posttranslational Modifications of Proteins in Type 1 Diabetes: The Next Step in Finding the Cure? <https://diabetes.diabetesjournals.org/content/61/8/1907>

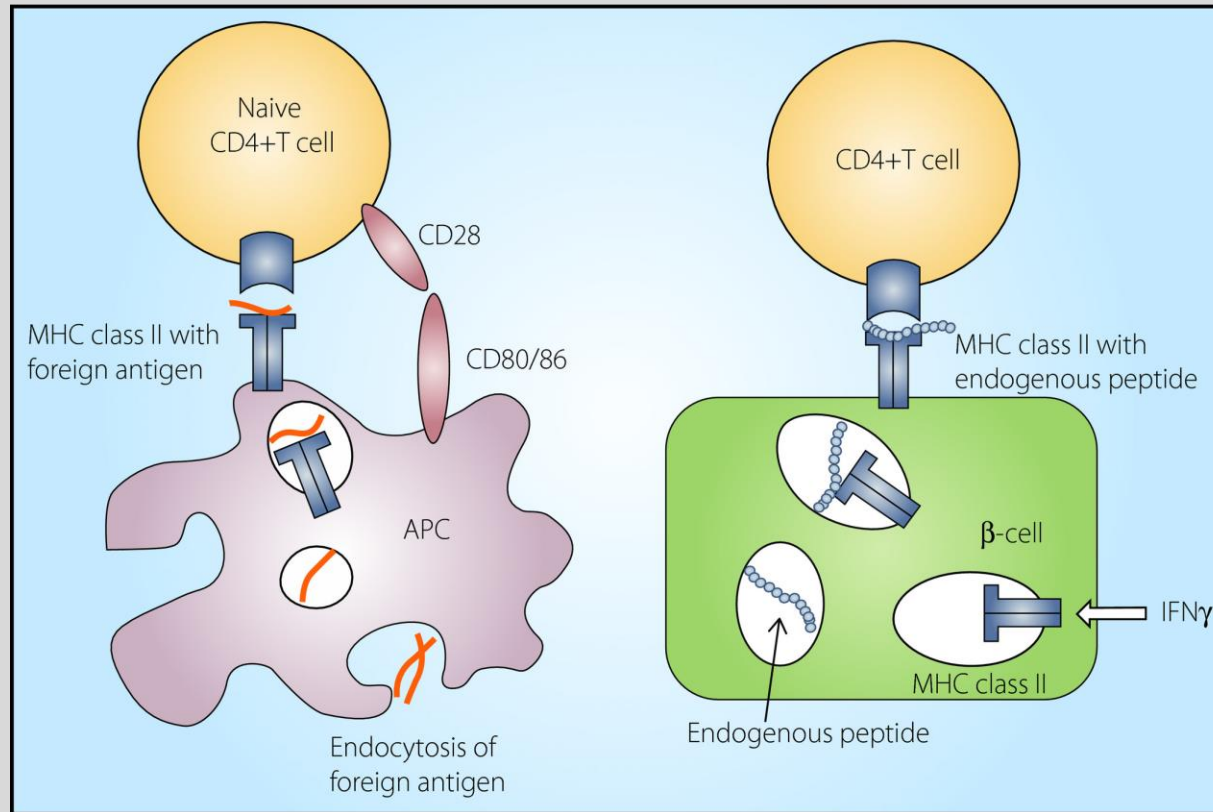
- **Hapten Effect**

- Host antigen becomes incorporated into an infection virion → induction of autoimmune response
- Heparin-induced thrombocytopenia: PF4/heparin complex



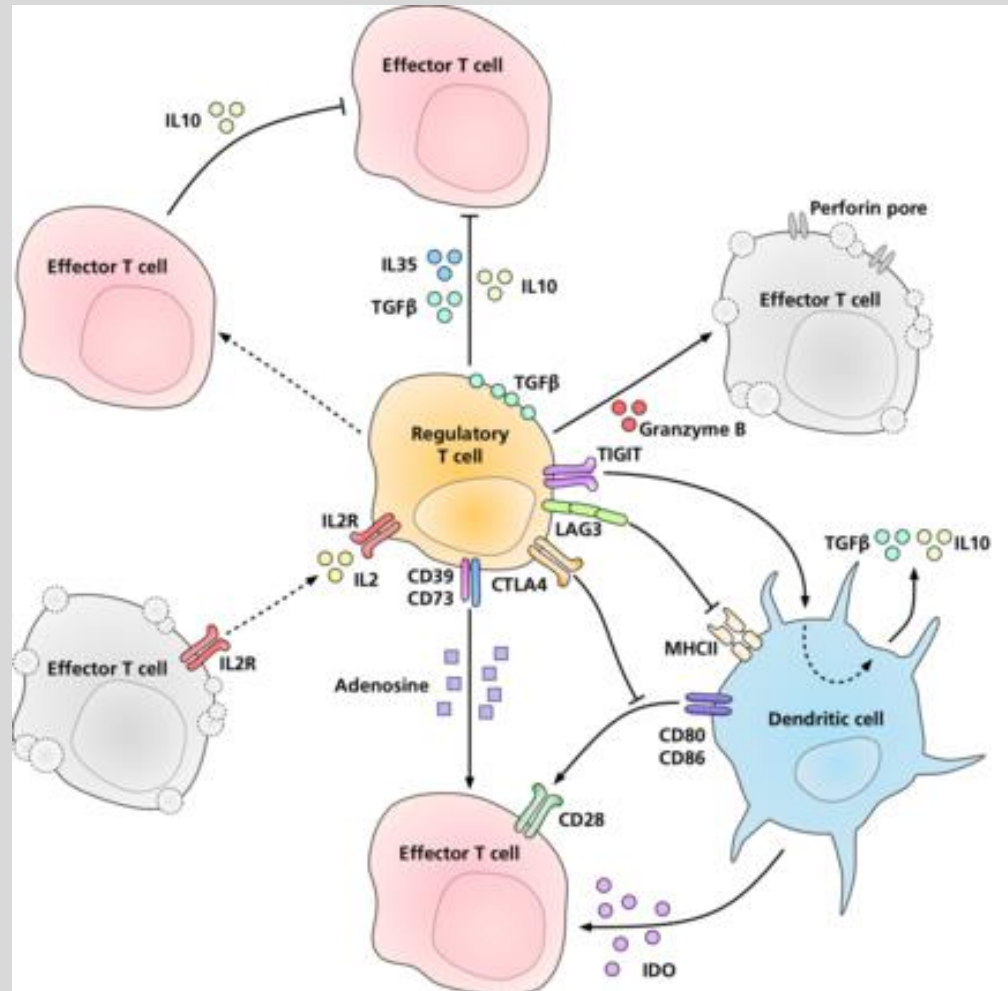
Source: https://www.uptodate.com/contents/image?topicKey=90261&search=HIT&source=outline_link&imageKey=HEME%2F50473

Inappropriate MHC Expression and Upregulation



Source: <https://onlinelibrary.wiley.com/doi/full/10.1111/jdi.13155>

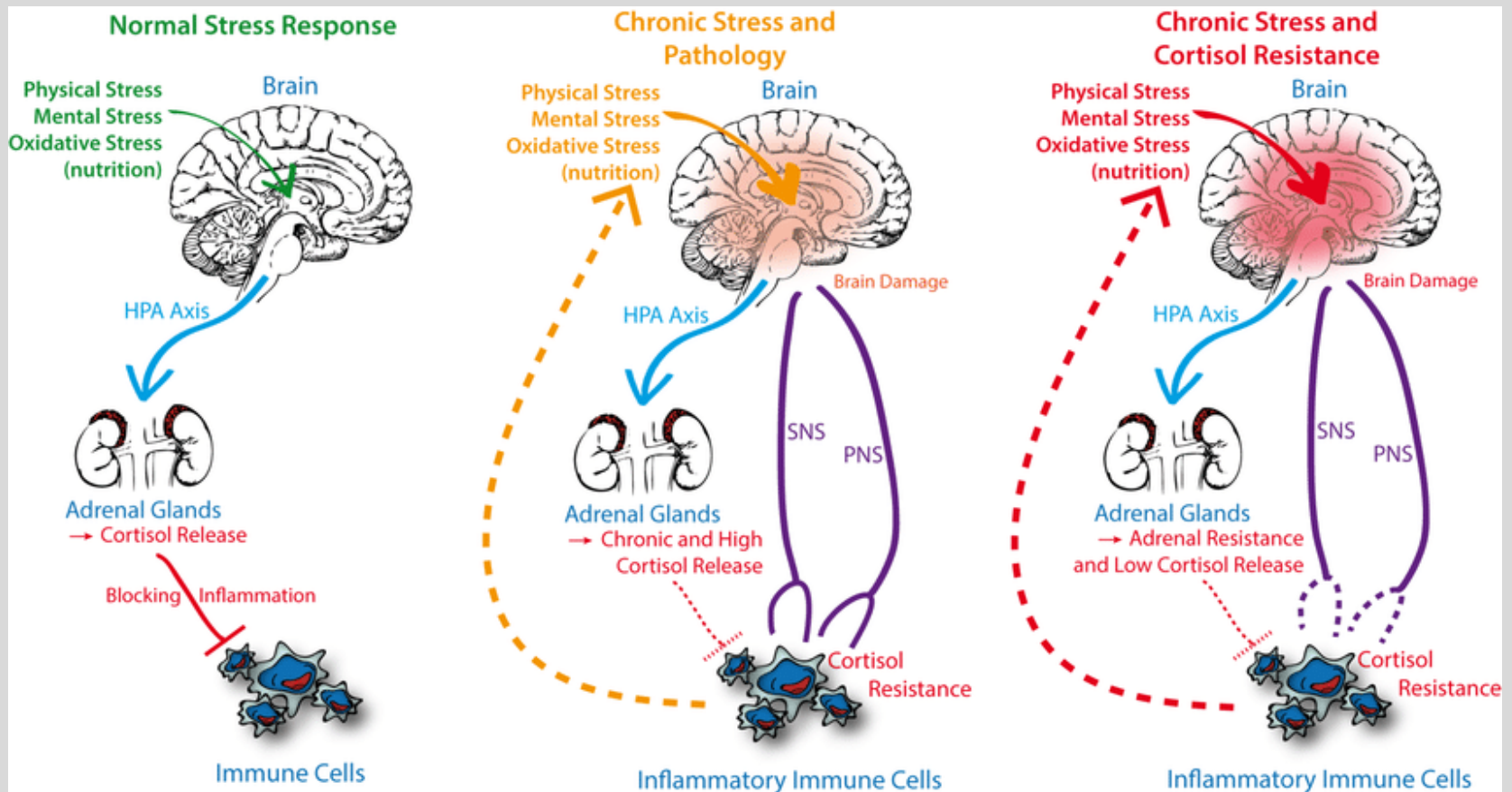
Defective T_{reg} Response



Source: https://www.wikiwand.com/en/Regulatory_T_cell

• Hormonal Imbalances

- Low circulating corticosteroid levels

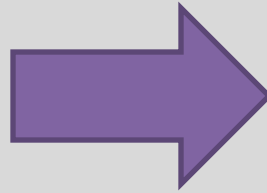


Cytokine Dysregulation

excessive cytokine
production

or

defective cytokine
receptors



aberrant immune
response and/or
activation of
anergic self-
reactive T-cells

*(IL2RA; TNF-, INF γ
and B-cell destruction)*

- **Pre-existing defects in the target organ:**
 - a) Increase uptake of iodine by thyroid
 - b) Risk factors linked to a TF controlling the rate of insulin production in type I diabetes

Review Questions

The _____ of *Streptococcus pyogenes* is similar to
_____.

This can lead to _____ and is an example of
_____.

- Tolerance is a complex immunological process that occurs in primary and secondary lymphoid organs
- Tolerance rupture → autoimmunity
- Autoimmunity: 2 clinical categories
- Autoimmune disease involves 3 stages
- Mechanisms/ongoing research

Core References:

Chapter 11 and 20 in D. Male, R. Stokes Peebles, and V. Male. 2021. *Immunology* 9th edition. Elsevier publisher. ISBN: 978-0-7020-7844-6

Chapter 4 in SG.

West and J. Kolfenbach. 2020. *Rheumatology Secrets*. 4th edition. Mosby, Inc., an imprint of Elsevier Inc. ISBN: 978-0-323-64186-9

Thank you

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>